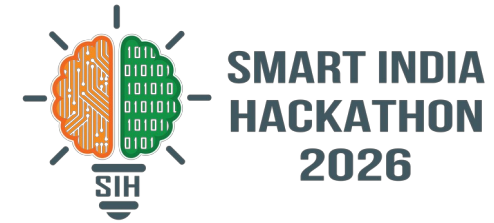


The logo for the Smart India Hackathon (SIH) is centered. It features a stylized brain shape composed of two halves: the left half is orange with white circuitry patterns, and the right half is green with white binary code (0s and 1s). Below the brain is a dark blue horizontal bar with the letters 'SIH' in white. The entire logo is set against a light gray background with a large hexagonal shape behind it.

- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Team ID-**
- **Team Name (Registered on portal)**

CVASSURE — INTEGRITY YOU CAN PROVE



Prove the data, the model and the answer — offline, for computer-vision pipelines fed by many contributors · PS 26228

● Proposed Solution

Three questions, answered before anything goes into service:

1. Was the training data tampered with — and **who sent it**?
2. Is this the model we were given, or has it been **swapped or edited**?
3. Is the record of what the model answered **trustworthy**?

```
cvassure audit --dataset data/ --model m.onnx --out results/
```

One command → one self-contained HTML report, a traffic-light verdict, and a coverage table that also lists what we **cannot** detect.

● How it addresses the problem

- ▶ **Names the source, not 900 files.** Beta-Binomial risk per contributor — an order an officer can act on.
- ▶ **Works black-box.** Detects a swapped model from its answers alone; refuses checks above the tier.
- ▶ **Proof for the log.** Signed, hash-chained receipts — cryptography, not statistics.

● Innovation and uniqueness

- ▶ **Plain English, enforced in code.** A finding with jargon or no number is rejected at construction.
- ▶ **The detectors cannot see the answer key.** A test reads the source and fails the build if they could.
- ▶ **It publishes its own weaknesses.** Coverage is generated from measurements, never hand-written.

1.00

duplicate floods &
foreign images caught

0.93

pasted markers &
label flips caught

100%

log tampering caught,
right failure named

0.20

blended triggers — we
report this as a gap

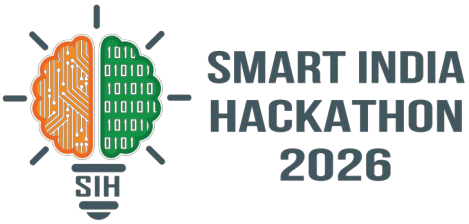
273

tests · 0 network calls
225 measured cells

Measured over 225 swept cells (5 attacks × 5 poison rates × 3 access tiers × 3 seeds) plus 12 tampering attempts — graded on the median cell, never the best.



TECHNICAL APPROACH



Technologies to be used

Core

Python 3.11 · NumPy · SciPy · scikit-learn · Pillow

Vision

PyTorch / TorchScript · ONNX Runtime · bundled ResNet-18

Crypto

Ed25519 (cryptography) · SHA-256 hash chains · pHash

Formats

COCO · YOLO · ImageFolder | ONNX · TorchScript

Output

Matplotlib · Jinja2 · single-file offline HTML

Access tiers — declared, and enforced

Tier 0

answers only → fingerprint + every data check

Tier 1

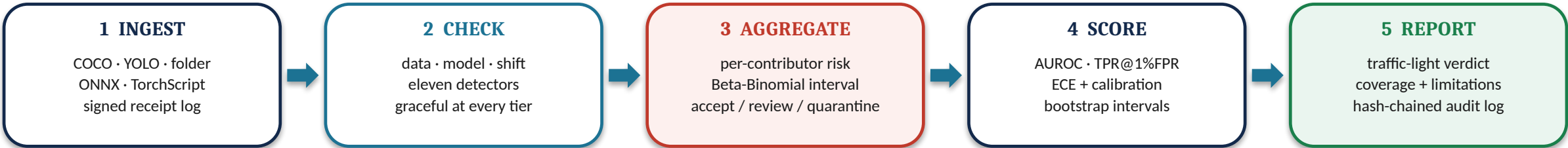
+ weights → per-layer digest, weight statistics

Tier 2

+ internals → spectral signature, trigger reconstruction

The handle refuses anything above the declared tier — even when the file format would allow it.

Methodology — one pass over the intake, then one report



Validated blind: a seeded attack harness poisons the data and keeps the answer key in a separate folder. **detect/ cannot import attack/** — a test reads the source and fails the build if it ever could. No retraining anywhere in the audit (PS 2.2.6) · `--offline-assert` replaces the socket layer and fails the run if anything reaches the network · encoder weights bundled in the repo.

The eleven checks, and the tier each one needs

near_duplicate

ood

label_noise

trigger_freq

contributor

shift

fingerprint t0

weight_digest t1

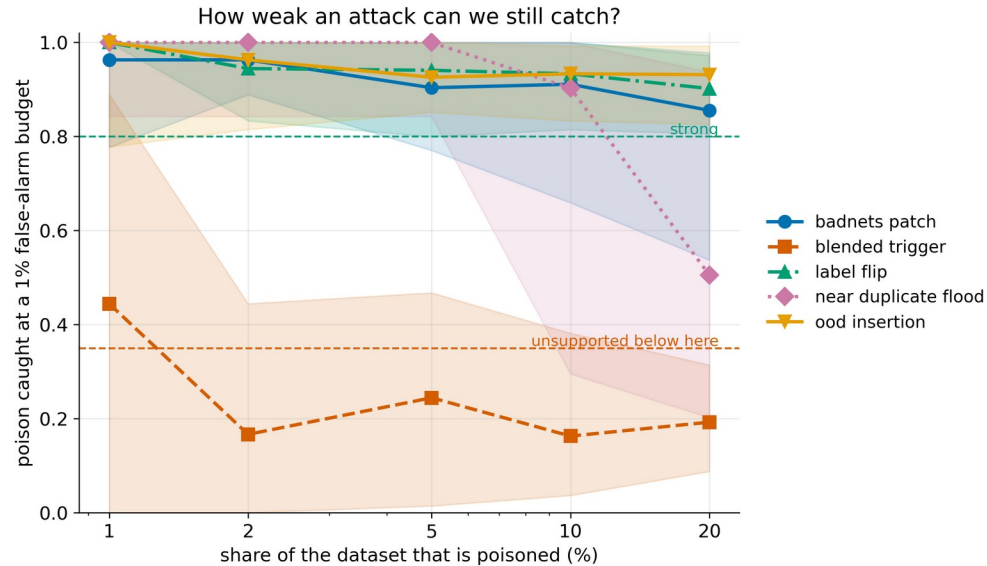
weight_stats t1

spectral_signature t2

trigger_recon t2

Every one degrades to a clean 'we could not check this, and here is why' at a lower tier — never a crash, never an invented number. A test runs the whole suite at tier 0 and asserts zero crashes.

● Measured, not asserted — 225 cells, 3 seeds each



Four attacks hold above 'strong' down to 1% poison. Blended triggers do not — and we say so.

● Measured coverage — graded on the median cell, never the best

What an attacker did	Typical TPR@1%FPR	Range	False alarms	Status
The same photograph submitted many times	1.00	0.45 – 1.00	3.0%	strong
Images from a completely different source	1.00	0.73 – 1.00	1.3%	strong
A marker pasted onto the image	0.93	0.67 – 1.00	1.7%	strong
The picture is fine, the label is wrong	0.93	0.83 – 1.00	1.0%	strong
A faint pattern over the whole image	0.20	0.00 – 0.67	1.1%	UNSUPPORTED
Any tampering with the inference log	1.00	proof, not an estimate	0.0%	strong

● Analysis of the feasibility

- ✓ **Already built and measured.** Not a concept — 273 tests, 225 swept cells, four results tables.
- ✓ **Commodity hardware.** 347 images audited in 13 s on a laptop CPU. No GPU, no cluster, no cloud.
- ✓ **Deploys into an air-gap.** Encoder weights ship inside the repository; nothing is downloaded.

● Potential challenges and risks → strategies

- ! **Faint blended triggers evade us (0.20).**
 - Reported as unsupported, not hidden. Tier-2 spectral check is the next build.
- ! **No contributor metadata in the intake.**
 - Regex fallback on paths; otherwise the report says so instead of guessing.
- ! **Vendor supplies a black box.**
 - Fingerprinting still proves substitution at tier 0; coverage marks the rest.
- ! **An attacker who knows our checks.**
 - Stated in every report. The detector set is pluggable and instantly measurable.

IMPACT AND BENEFITS

What an analyst actually receives

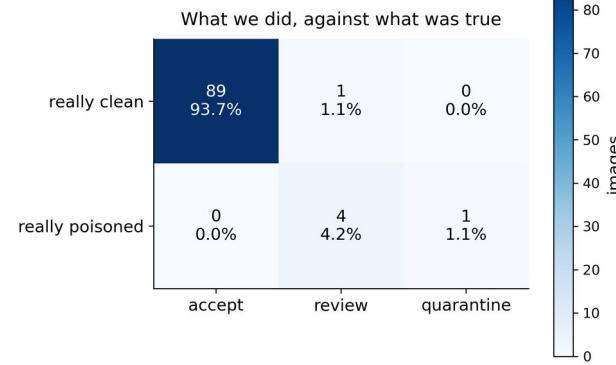
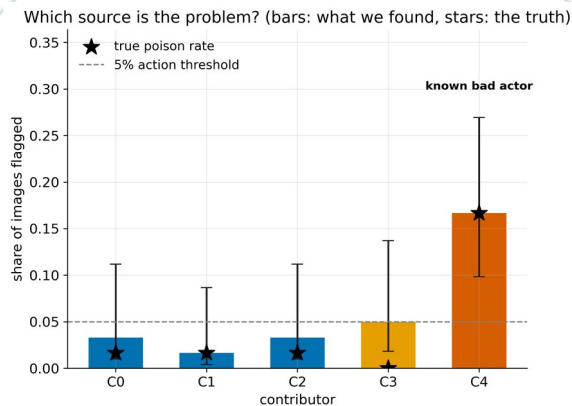
QUARANTINE RECOMMENDED

1 of 5 contributors clean. 3 contributors quarantined (C2, C3, C4). 1 to review.

Model: SUBSTITUTED OR EDITED.

Inference log: 2 records tampered (#347, #348).

Real output. One offline HTML file — no dashboard to host, nothing to install on the analyst's machine.



Before: an analyst is handed 900 suspect image IDs and no way to rank them.

After: one line — quarantine Contributor C3, whose true poison rate was 51% and whom we scored 45% with a 36–54% interval.

Potential impact on the target audience

- ▶ **Intake officer** gets an order — 'quarantine Contributor D' — not 900 IDs to triage.
- ▶ **Model reviewer** can prove a vendor model is the enrolled one, from its answers alone.
- ▶ **Investigator** gets a named failure mode on the exact record that was edited.
- ▶ **Commander** sees one traffic light, and an honest list of what was not checked.

Benefits of the solution

Operational · Triage collapses from every image to a handful of sources — minutes, not days.

Security · Poisoned data and swapped models are caught before they reach a model that guides a decision.

Legal / audit · Every inference is provable after the fact; the audit trail is itself tamper-evident.

Economic · Contributions are salvaged, not binned — quarantine one source instead of discarding a whole intake.

Ethical · The system states its own limits in every report, so nobody over-trusts a clean result.

RESEARCH AND REFERENCES

● Attack literature we test against

BadNets — Gu, Dolan-Gavitt & Garg, 2017. Patch triggers. arXiv:1708.06733

Blended / invisible triggers — Chen et al., 2017. The attack we measure worst on. arXiv:1712.05526

Spectral Signatures — Tran, Li & Madry, NeurIPS 2018. Our tier-2 check. arXiv:1811.00636

Neural Cleanse — Wang et al., IEEE S&P 2019. Trigger reconstruction, MAD anomaly index.

Dataset poisoning survey — Goldblum et al., TPAMI 2023. Multi-contributor threat model.

● Methods and standards

Ed25519 — RFC 8032. Keys generated locally, never transmitted.

Isotonic calibration — Zadrozny & Elkan, KDD 2002. Score → probability.

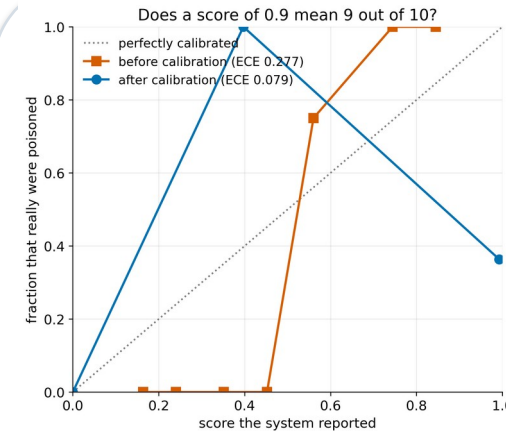
Benjamini-Hochberg — JRSS-B 1995. Controls false alarms across the nine shift measurements.

COCO / YOLO — Lin et al., ECCV 2014; Ultralytics data.yaml. Both ingest formats PS 2.2.6 requires.

● Datasets

SYNTH-10 — procedural, ships with the repo so every result reproduces air-gapped · CIFAR-10 · GTSRB · VisDrone / DOTA subset, in both COCO and YOLO form.

● Our own measured results



Calibration works. Expected Calibration Error falls from 0.277 to 0.079 once the combined score is calibrated as a quantity in its own right.

Sanity-checked first. Random scores must give AUROC 0.5, perfect scores 1.0 — both are tests, written before any detector existed.

Every headline number carries a bootstrap 95% interval; every swept cell runs at three seeds.

● Repository and reproduction

make reproduce regenerates every number and figure from scratch, offline.

make verify runs 273 tests plus the assertion that nothing touched the network.

make demo the four-minute judge sequence, laptop in airplane mode.

Full source, the seeded attack harness and all four results tables ship in the repository.

PS 26228 covered:

✓ 2.2.1 data integrity

✓ 2.2.2 model integrity

✓ 2.2.3 provenance

✓ 2.2.4 calibration + shift

✓ 2.2.5 analyst report

✓ 2.2.6 formats · tiers · offline